12.8 Solids of Revolution

Name _____ Period ____ Date _____

Geometry 12.8 (30-minute plan) · Student sheet

I can…

- name the solid formed by rotating a figure around an axis.
- tell which measures become the radius and the height.
- find the volume of a solid of revolution.

Vocabulary · fill in the missing word

Solid of revolution — a _____ figure formed by rotating a two-dimensional shape around an axis

Axis of revolution — the line around which the shape is _____

Part 1 · Notes and examples (fill in each note, then work the example)

Basic solids of revolution: a rectangle rotated around one side makes a _____; a right triangle rotated around a leg makes a _____; a semicircle rotated around its diameter makes a _____.

Example 1

Name the solid formed by rotating each figure around the given axis.

a. a rectangle around one of its sides

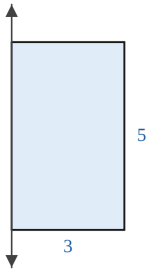
b. a right triangle around one of its legs

c. a semicircle around its diameter

Radius and height: the side on the axis becomes the _____; the side perpendicular to the axis becomes the _____.

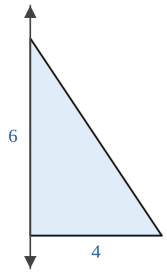
Example 2

The rectangle is rotated around the side on the axis. Find the volume of the solid. Round to the nearest hundredth.



Example 3

A right triangle with legs of 6 (on the axis) and 4 is rotated around the axis. Find the volume of the solid. Round to the nearest hundredth.



Part 2 · Partner practice

Partner practice

A semicircle with a radius of 3 is rotated around its diameter. Find the volume of the solid. Round to the nearest hundredth.